



# PROJECTPLAN JUGO

GROUP 8 AMD

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**Stakeholder**

Jugo

**Stakeholders**

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**Project Coach**

Senne

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## 1. Project Context

### 1.1 Project Description

The goal of this project is to design and develop an **AI-driven application** that evaluates data visualizations for compliance with **IBCS (International Business Communication Standards)**.

Many dashboards and data visualizations are created without following IBCS guidelines. This reduces clarity, consistency and interpretability of business information.

This project aims to create an AI-based system that can automatically analyze dashboard images and determine whether they follow specific IBCS design rules. If non-compliance is detected, the system will provide feedback and suggestions for improvement.

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### 1.2 Project Goal

The goal of the project is to develop an AI application that:

- Analyzes dashboard images
  - Detects whether IBCS rules are followed
  - Identifies visual design issues
  - Provides recommendations to improve compliance
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### 1.3 Final Product

The final product will be a **web-based AI application**.

Users will be able to:

- Upload dashboard images
  - Receive an AI-based evaluation
  - Receive suggestions to improve IBCS compliance
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## **2. AI / Machine Learning Approach**

### **2.1 Machine Learning Goal**

The machine learning system will classify whether dashboard visualizations comply with specific IBCS rules.

This means the model will analyze visual patterns in dashboard images and determine compliance.

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### **2.2 Target Variable**

The target variable is:

**IBCS compliance score or classification of a dashboard visualization**

Possible outputs include:

- Compliant
  - Non-compliant
  - Compliance score
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### **2.3 Data Type**

The project will use **image data**.

The dataset will consist of dashboard images and data visualization screenshots.

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### **2.4 Data Collection**

The dataset will be created by:

- Scraping dashboards from online sources
- Collecting publicly available dashboards
- Creating dashboards manually if needed

Manual creation may be required to ensure sufficient examples of compliant and non-compliant visualizations.

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### 3. System Architecture

The application will consist of three main components.

#### Frontend

React web application

Responsible for:

- Uploading dashboard images
  - Displaying AI results
  - Showing feedback and suggestions
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#### Backend

Python backend

Responsible for:

- Handling API requests
  - Running the machine learning model
  - Processing images
  - Returning predictions
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#### Machine Learning Model in Python

Responsible for:

- Image analysis
  - Feature extraction
  - Compliance classification
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#### **4. Development Stack**

##### **Frontend**

React

##### **Backend**

Python

##### **Development Environment**

VSCode

##### **Version Control**

GitHub

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##### **Tools Used**

The following tools will be used during development.

VSCode: development IDE

GitHub: code

Trello: planning

Microsoft Teams / Onedrive: documents

WhatsApp: communication

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## **5. Team Members**

Ayoub Lafhailat  
Dimitar Malamski  
Ruben Wassenberg  
Ali Al-Bukhaiti  
Iris Prooijen  
Nahia Ortiz Fandiño

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## **6. Team Roles**

### **Project Leader**

Nahia

Responsible for:

- Project coordination
  - Communication between team members
  - Solving project issues
  - Monitoring project progress
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### **Scrum Master**

Dimitar

Responsible for:

- Sprint planning
  - Maintaining Trello board
  - Managing sprint workflow
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### **Meeting Notes**

Ruben

Responsible for documenting meeting outcomes.

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### **Feedpulse**

Ruben

Responsible for maintaining Feedpulse documentation.

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### **Communication with Stakeholders and Teachers**

Ali

Responsible for communication with stakeholders and teachers.

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### **Code Reviewers**

Ayoub and Dimitar

Responsible for reviewing pull requests before merging.

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### **Presentation Preparation**

Iris

Responsible for preparing presentations and demos.

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### **Coding Quality Documentation**

Ayoub

Responsible for creating coding guidelines.

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### **GitHub Workflow Management**

Dimitar

Responsible for defining GitHub rules.

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## 7. Scrum Workflow

The project will follow an **Agile Scrum workflow**.

### Sprint Duration

4 weeks

### Number of Sprints

5

Each sprint will focus on a specific stage of development.

## 8. Sprint Planning

### Sprint Focus

Sprint 1 Domain understanding & research

Sprint 2 Data collection & dataset creation

Sprint 3 Model development

Sprint 4 Model improvement & evaluation

Sprint 5 Web application integration & final demo

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## 9. Trello Workflow

The team will use Trello to manage tasks.

### Task Columns

To Do

In Progress

Review

Done

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### Task Rules

- Every task must be created in Trello
  - Each task must have an assigned team member
  - Tasks must have a clear description
  - Tasks should be small enough to complete within 1–3 days with a clear definition of done
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## 10. GitHub Workflow

All code will be stored in a shared GitHub repository.

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### Branching Strategy

Main branch

Stable production-ready code.

Development branch

Integration branch for new features.

Feature branches

Created for individual tasks.

Example:

feature/data-collection

feature/model-training

feature/frontend-upload

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### Pull Request Rules

- No direct commits to main
  - All changes must go through pull requests
  - At least one code reviewer must approve the pull request
  - Code must run without errors before merging
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## 11. Coding Standards

To maintain code quality the following standards will be used.

### Python

- Use meaningful variable names
  - Write clear function names
  - Follow OOP style guidelines
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### React

- Use component-based structure
- Keep components small and reusable
- Separate UI logic and data logic

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## 12. Meetings

### Weekly Standups

Tuesday – 10:00

These meetings are held **physically (not online)**.

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### Stakeholder Meetings

Stakeholder meetings take place **monthly**.

All team members attend.

Two members present the demo.

Presenters rotate every meeting.

Other team members answer technical questions.

## 13. Demo Structure

Each demo lasts **30 minutes**.

Structure:

1. Product demonstration
2. Technical explanation
3. Planning next sprint
4. Discussion and feedback

## 14. Communication

Communication tools:

WhatsApp – quick communication

Teams – document sharing

GitHub – code collaboration

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## 15. Definition of Done

A task is considered finished when:

- Code runs without errors
- Code is pushed to GitHub
- Pull request is approved
- Documentation is updated
- Task is marked done in Trello

## 16. Risk Analysis

### Stakeholder Response Delays

Stakeholder responses may take **1–2 days or longer**.

Mitigation:

- Prepare questions in advance
- Communicate clearly during meetings

### Dataset Availability

Obtaining enough IBCS-compliant dashboard images may be difficult.

Mitigation:

- Create dashboards manually
- Generate additional examples if necessary

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## 17. Collaboration Rules

To ensure effective collaboration:

- Respect deadlines
- Communicate openly
- Help teammates when needed
- Inform the team if you cannot attend a meeting at least **24 hours in advance**
- Inform team members timely when your stuck on a stuck